Beyond Item Selection: Rethinking Ability Evolution in Computerized Adaptive Testing

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Abstract

Computerized Adaptive Testing (CAT), as a core assessment technique in intelligent education, aims to enhance the efficiency and precision of learner evaluation by adaptively selecting questions that best align with the examinee's current knowledge state. Indeed, recent years have witnessed substantial progress in the design and refinement of the fundamental cognitive diagnosis models and the question selection strategies, leading to promising results. However, the evolutionary dynamics of examinee abilities and the underlying updating mechanisms throughout the adaptive testing process have received limited attention. To this end, this paper investigates the impact of update mechanisms with varying granularities on the accuracy of ability estimation, with a particular focus on step-based and batch-based strategies in the context of assessment diagnosis. Specifically, we begin by conducting an approximation analysis to compare the changes in ability estimates induced by the two strategies, followed by a theoretical analysis of their update trajectories through the lens of error bounds. To trade-off the adaptability and stability during the assessment process, we propose **SUAT**, a **S**mooth-step **U**pdating strategy for more effective **A**daptive **T**esting. Finally, we conduct extensive experiments on two real-world educational datasets of varying scales, and provide a detailed analysis and discussion of the differences among the different update mechanisms, and their potential applications.

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KDD '26, Jeju Island, Republic of Korea
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ACM ISBN 979-8-4007-2259-2/2026/08
https://doi.org/10.1145/3770855.3817615

CCS Concepts

• **Applied computing** → **E-learning**; **Learning management systems**; • **Information systems** → **Personalization**.

Keywords

Educational data mining, knowledge management, adaptive testing, learning item selection, ability update strategy

ACM Reference Format:
Xiaoshan Yu, Jian Li, Shangshang Yang, Ziwen Wang, Haiping Ma, and Xingyi Zhang. 2026. Beyond Item Selection: Rethinking Ability Evolution in Computerized Adaptive Testing. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26), August 09–13, 2026, Jeju Island, Republic of Korea.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3770855.3817615

1 Introduction

With the advancement of intelligent education [6, 29], personalized learning [13] has emerged as a prominent educational paradigm, offering tailored learning services that align with individual learner needs and enhance the efficiency of goal attainment. Computerized adaptive testing (CAT) [15], as a core and promising assessment technique in personalized learning, aims to accurately evaluate learners' knowledge levels by dynamically selecting testing items tailored to their abilities, and has attracted increasing attention in recent years, especially in web-based online education systems [7].

In general, CAT comprises two core modules [35]: a question selection module $\mathcal{M}_s$, which interactively selects the most appropriate question item at each step, and an ability assessment module $\mathcal{M}_a$, which continuously diagnoses the test taker's knowledge proficiency based on real-time feedback, as illustrated in Figure 1(a); these two modules operate alternately to enable adaptive testing

until a predefined termination criterion is met. Owing to its inherently adaptive nature [3, 26], most existing research has focused on optimizing the question selection module [22, 48, 55], which plays a central role in determining the effectiveness of CAT and is broadly approached through either rule-based or data-driven methods.

Representing the early stage of CAT development, rule-based approaches [1, 4, 17] commonly rely on heuristic strategies to select items that align with the examinee's current ability or provide high diagnostic value. For example, MAAT [1] quantifies item informativeness using the expected maximum change criterion, a principle drawn from active learning [34]. In contrast, data-driven methods [12, 56] have emerged as a powerful alternative by leveraging learner interaction data to learn adaptive item selection strategies. For instance, BOBCAT [12] formulates CAT as a bi-level optimization problem, allowing the item selector to be learned from data and avoiding the rigid assumptions of traditional approaches.

Despite the promising results of existing methods, limited attention has been paid to the assessment and evolution of examinee abilities throughout the adaptive testing process, particularly regarding the influence of different update mechanisms on ability trajectories. As shown in Figure 1(b), the CAT process involves continuous interaction between the test taker and the test items, requiring real-time feedback from the diagnosis model. This feedback is realized by dynamically updating the test taker's ability parameters within the model $\mathcal{M}_a$. Notably, the effectiveness of step-wise versus batch updates in estimating final abilities can diverge significantly, especially under varying adaptive testing lengths, diagnosis models, or question selection strategies, as demonstrated by the contrasting trajectories of the red curve and green line in the figure. Intuitively, the step-based strategy provides immediate feedback and finer-grained updates, but are more susceptible to volatility, potentially leading to sensitivity in the results. In contrast, the batch-based strategy updates less frequently but make more intensive use of accumulated samples. Investigating the impact of these two strategies on the evolution of proficiency estimation can offer valuable insights for improving item selection and basic diagnosis.

To this end, unlike traditional research on CAT, this paper focuses on investigating the effects of different update mechanisms on examinee ability estimation and their potential implications for the adaptive testing process, as illustrated in Figure 1(c). Specifically, we begin by examining the update processes of step-based and batch-based strategies separately, and theoretically explore the similarity of their ability updates from the perspective of update intensity. We further analyze the differences in their update trajectories by deriving error bounds. To address the trade-off between adaptability and stability during the evaluation process, we propose **SUAT**, a **S**mooth-step **U**pdating strategy for more effective **A**daptive **T**esting. Finally, we conduct extensive experiments on two educational datasets of varying scales, and provide a detailed analysis and discussion of the differences among the update mechanisms, the insights they reveal, and their potential applications.

2 Related Work

2.1 Computerized Adaptive Testing

Computerized adaptive testing (CAT) [24], a foundational component of intelligent education [19, 29, 50, 54], emerges from the intersection of computer science and educational assessment. Its core objective is to precisely diagnose and evaluate examinees' knowledge proficiency by adaptively selecting learning items in a personalized and efficient manner. Unlike traditional paper-and-pencil tests [23, 27, 41, 53], CAT aims to dynamically adjust question selection based on individual responses, offering a fine-grained and tailored assessment of each learner's ability [21, 46, 51]. Recent studies have primarily focused on optimizing the question selection mechanism [49], which is central to the effectiveness of CAT and can be broadly divided into two categories [15]: rule-based and data-driven approaches.

Rule-based approaches [1, 4, 16, 17], which characterize the early stage of CAT development, typically adopt heuristic strategies to select items that either match the examinee's current proficiency or offer high diagnostic utility [45, 47]. For instance, BECAT [57] selects items by approximating full-response gradients, achieving accurate ability estimates with fewer questions and theoretical error guarantees. On the other hand, data-driven methods [12, 37, 56] have gained prominence by learning adaptive item selection strategies directly from learner interaction data. For example, BOBCAT [12] formulates CAT as a bi-level optimization problem, where the item selector is learned from data, thereby overcoming the rigidity and assumptions of traditional rule-based or IRT-based models. Despite their notable success in optimizing item selection, these methods typically default to a step-based update scheme for ability estimation during the testing process. However, little attention has been paid to the impact of different update mechanisms on assessment dynamics [9, 20, 33, 38], which motivates this work.

2.2 Latent Trait Estimation

Latent trait estimation [8, 14, 31, 40, 43], which aims to infer unobservable student abilities based on their responses to assessment items, lies at the core of cognitive diagnostic modeling. Early approaches are rooted in Item Response Theory (IRT) [10], where each student is characterized by a continuous latent trait θ_i, and the probability of answering item e_j correctly is modeled as: $P(X_{ij} = 1 \mid \theta_i) = \sigma(a_j \cdot (\theta_i - \beta_j))$, where a_j and β_j denote the discrimination and difficulty parameters of item e_j, respectively, and $\sigma(\cdot)$ is the logistic sigmoid function. VIRT [40] introduces a fast and scalable variational Bayesian inference method for Item Response Theory, enabling accurate estimation and expressive extensions for improved prediction and missing data imputation on large-scale datasets. While effective for global ability estimation, IRT lacks the granularity needed to capture students' specific strengths and weaknesses on fine-grained skills. Recent advances have extended cognitive diagnosis models (CDMs) with neural architectures [11, 18, 36, 44, 52], enabling the learning of continuous latent representations for students and items. These models [39, 42] replace manual Q-matrix design with learned embeddings, offering more flexible and expressive latent trait estimation while preserving educational interpretability. For example, RCD [11] enhances latent trait estimation by modeling multi-level relations among students, questions, and concepts, capturing both structural and interactive dependencies critical for accurate ability inference. However, limited research has focused on how the assessment of students' latent traits evolves throughout the adaptive testing process, particularly

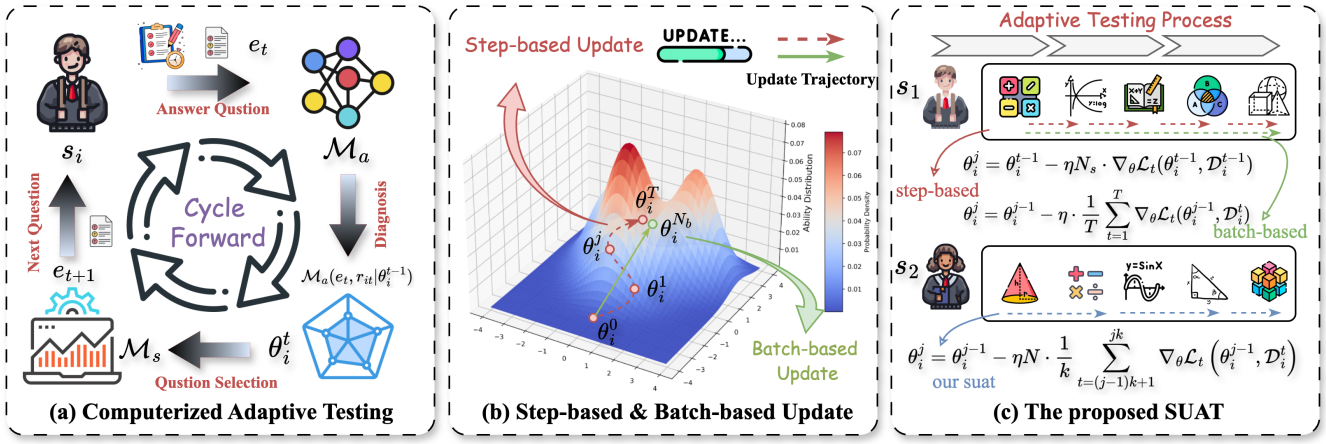


Figure 1: (a) Flowchart of computerized adaptive testing (CAT), mainly including a diagnosis model $\mathcal{M}_s$ and a selection model $\mathcal{M}_a$; (b) Visualization of the update trajectories; (c) Comparison of different update strategies and the proposed SUAT.

with respect to different update scales, and this gap forms the foundation of our research.

3 Methodology

3.1 Problem Definition

The goal of computerized adaptive testing (CAT) is to deliver tailored exercises or testlets to examinees in an autoregressive manner, with the aim of efficiently evaluating their corresponding knowledge levels. Let $\mathcal{S} = \{s_1, \ldots, s_{|\mathcal{S}|}\}$ be the set of examinees, $\mathcal{E} = \{e_1, \ldots, e_{|\mathcal{E}|}\}$ be the question bank, and $\mathcal{C} = \{c_1, \ldots, c_{|\mathcal{C}|}\}$ be the set of knowledge concepts. The association between questions and knowledge concepts is typically captured using a Q-matrix, denoted as $\mathbf{Q} = \{q_{ij}\}^{|\mathcal{E}|\times|\mathcal{C}|}$, where $q_{ij} = 1$ indicates that question e_i is related to concept c_j, and $q_{ij} = 0$ otherwise. For each examinee $s_i \in \mathcal{S}$, the interaction records is denoted by $\mathcal{R}_i = \{(s_i, e_j, r_{ij}) \mid e_j \in \mathcal{E}, r_{ij} \in \{0, 1\}\}$, where $r_{ij} = 1$ indicates that the examinee s_i answered the question e_j correctly, while $r_{ij} = 0$ signifies an incorrect response. As previously mentioned, CAT systems typically consist of two core components: **(1)** a *question selection module* $\mathcal{M}_s$ that iteratively selects the most appropriate questions for the examinee, and **(2)** an *ability assessment module* $\mathcal{M}_a$ that accurately evaluates the examinee's current proficiency based on interactive response data. Therefore, each step of the adaptive testing process can be represented as follows:

$$\begin{aligned} e_t &\leftarrow \mathcal{M}_s(\mathcal{E}_i \mid \theta_i^{t-1}), \\ \theta_i^t &\leftarrow \mathcal{M}_a(e_t, r_{it} \mid \theta_i^{t-1}), \end{aligned} \tag{1}$$

where e_t denotes the exercise selected by $\mathcal{M}_s$ from the question bank of the student s_i based on his/her prior ability θ_i^{t-1}, and θ_i^t represents the examinee's current ability, updated by the evaluation model $\mathcal{M}_a$ following the t-step interaction (e_t, r_{it}), After T rounds of interaction, the examinee's final estimated ability θ_i^T and the corresponding exercising trajectory $\mathcal{D}_i^T = \{(e_1, r_{i1}), \ldots, (e_T, r_{iT})\}$ are obtained. Formally, the overall optimization objective of the CAT task involving $\mathcal{M}_s$ and $\mathcal{M}_a$ can be defined as follows:

$$\begin{cases} \mathcal{M}_s^* = \arg\min\limits_{\mathcal{M}_s} \dfrac{1}{|\mathcal{S}|} \sum\limits_{i=1}^{|\mathcal{S}|} \sum\limits_{t=1}^{T} \dfrac{1}{|\mathcal{D}_i'|} \sum\limits_{(e,r)}^{\mathcal{D}_i'} \mathcal{L}_t(r, \mathcal{M}_a(e|\hat{\theta}_i^t)), \\ \text{s.t.} \quad \hat{\theta}_i^t = \arg\min\limits_{\theta_i^t} \sum\limits_{(e_j, r_{ij})}^{\mathcal{D}_i^t} \mathcal{L}_t(r_{ij}, \mathcal{M}_a(e_j|\theta_i^t)), \\ \text{where} \quad \mathcal{D}_i' \cap \mathcal{D}_i = \emptyset, \\ \qquad \mathcal{D}_i^t = \{(e_1, r_{i1}), \ldots, (e_t, r_{it})\}, \quad \text{and} \\ \qquad e_t \sim \mathcal{M}_s((e_1, r_{i1}), \ldots, (e_{t-1}, r_{it-1})). \end{cases} \tag{2}$$

3.2 Update Approximate Analysis

Existing research on CAT [15] primarily centers on the design and improvement of $\mathcal{M}_s$ and $\mathcal{M}_a$, with particular emphasis on the item selection module due to its direct influence on assessment outcomes. However, the influence of different update strategies, especially step-based and batch-based updates, on the evolution of examinees' ability parameters during the testing process has received limited attention. The step-based update is more adaptive but noisier, while batch-based update is more stable but less responsive. Therefore, this section is dedicated to a in-depth exploration of ability evolution under the two strategies.

Specifically, for any examinee $s_i \in \mathcal{S}$, we assume a knowledge vector θ_i, initially set to θ_i^0. In the **step-based update** process, each step typically utilizes only the interaction data from the currently selected exercise to update the capability over N_s epochs, as follows:

$$\begin{aligned} \theta_i^t &= \theta_i^{t-1} - \eta \cdot \sum_{j=1}^{N_s} \nabla_\theta \mathcal{L}_t(\theta_i^{t-1}, \mathcal{D}_i^{t-1}), \\ &= \theta_i^{t-1} - \eta N_s \cdot \nabla_\theta \mathcal{L}_t(\theta_i^{t-1}, \mathcal{D}_i^{t-1}), \end{aligned} \tag{3}$$

where θ_i^t denotes the estimated ability at the t-th step, η is the learning rate (without decay), and $\mathcal{L}_t$ represents the interaction loss at step t. We assume that the loss function $\mathcal{L}(\cdot)$ is L-smooth and has a uniformly bounded gradient [30, 58]; that is, $\|\nabla\mathcal{L}(\theta)\| \leq G$

for all θ, and the gradient is L-Lipschitz continuous, satisfying $\|\nabla\mathcal{L}(\theta_1) - \nabla\mathcal{L}(\theta_2)\| \leq L\|\theta_1 - \theta_2\|$, therefore, after T interaction trajectory updates, the examinee's final estimated ability is given by

$$\theta_i^T = \theta_i^0 - \eta \sum_{t=1}^{T} N_s \cdot \nabla_\theta \mathcal{L}_t(\theta_i^{t-1}, \mathcal{D}_i^{t-1}). \tag{4}$$

In contrast, the **batch-based update** process adopts an offline strategy in which all items are pre-selected before testing. After completing the full T-step exercise interaction sequence, the model updates the ability parameters for N_b epochs using the entire trajectory information, achieving a more global optimization, as follows:

$$\theta_i^j = \theta_i^{j-1} - \eta \cdot \frac{1}{T} \sum_{t=1}^{T} \nabla_\theta \mathcal{L}_t(\theta_i^{j-1}, \mathcal{D}_i^t), \tag{5}$$

where θ_i^j represents the estimated ability at the j-th update step, with $j \in \{1, \ldots, N_b\}$. Each update utilizes the gradient information derived from all T interaction samples, thereby reducing estimation variance to a certain extent. Accordingly, the final ability obtained under the batch-based update strategy is expressed as follows:

$$\theta_i^{N_b} = \theta_i^0 - \eta \cdot \frac{1}{T} \sum_{j=1}^{N_b} \sum_{t=1}^{T} \nabla_\theta \mathcal{L}_t(\theta_i^{j-1}, \mathcal{D}_i^t). \tag{6}$$

To facilitate a more direct comparison of the approximation between the two update strategies in terms of student ability estimation, we analyze their behavior from the perspective of the Euclidean distance between their update trajectories. Specifically, the approximate total update magnitude for the step-based and batch-based strategies is computed using Eq. (4) and Eq. (6):

$$\begin{cases} \Delta_s \approx \Big\| \underbrace{-\eta \sum_{t=1}^{T}}_{\text{outer update}} \underbrace{N_s \cdot \nabla_\theta \mathcal{L}_t(\theta_i^{t-1}, \mathcal{D}_i^{t-1})}_{\text{inner update}} \Big\|, \\ \Delta_b \approx \Big\| \underbrace{-\eta \sum_{j=1}^{N_b}}_{\text{outer update}} \underbrace{\cdot \frac{1}{T} \sum_{t=1}^{T} \nabla_\theta \mathcal{L}_t(\theta_i^{j-1}, \mathcal{D}_i^t)}_{\text{inner update}} \Big\|, \end{cases} \tag{7}$$

where $\Delta_s = \left\|\theta_i^T - \theta_i^0\right\|$ and $\Delta_b = \left\|\theta_i^{N_b} - \theta_i^0\right\|$ denote the norms of the cumulative updates for the step-based and batch-based strategies, respectively. Accordingly, we then provide a detailed examination of the update approximation in the following discussion.

Theorem 3.1 (Ability Update Approximation). *Assume that the loss function $\mathcal{L}(\cdot)$ is differentiable with L-Lipschitz continuous gradients, and that the gradient norms remain uniformly stable throughout the update process. Let both the step-based and batch-based strategies start from the same initialization θ_i^0, and use a fixed learning rate η. Let Δ_s and Δ_b denote the cumulative update magnitudes under the step-based and batch-based strategies, respectively. Then, under the condition of comparable update magnitude, the relationship between their respective iteration counts satisfies:*

$$\Delta_s \approx \Delta_b \Longrightarrow N_b \approx T \cdot N_s. \tag{8}$$

Theorem 3.1 reveals the approximation between the two update strategies in terms of their influence on ability parameter evolution during adaptive interactions, thereby providing a theoretical foundation for ensuring the fairness of selection algorithm evaluation.

3.3 Trajectory Error Analysis

To further investigate the differences in the update trajectories of the two strategies from a theoretical standpoint, we conduct an analysis based on error bounds. Specifically, we consider that the current ability parameter θ_i^{t-1} at each update step is influenced by accumulated historical disturbances, which we denote as $\Delta_t = \nabla_\theta \mathcal{L}_t(\theta_i^{t-1}) - \nabla_\theta \mathcal{L}_t(\theta_i^0)$ (for brevity, we omit the data dependence $\mathcal{D}^t$ in $\mathcal{L}_t(\cdot)$). Under the L–smoothness assumption in Theorem 3.1, the corresponding Euclidean norm can be bounded as:

$$\|\Delta_t\| = \left\|\nabla_\theta \mathcal{L}_t(\theta_i^{t-1}) - \nabla_\theta \mathcal{L}_t(\theta_i^0)\right\| \leq L \cdot \left\|\theta_i^{t-1} - \theta_i^0\right\|, \tag{9}$$

where L denotes the Lipschitz constant, characterizing the upper bound of the function's gradient variation. By subsequently applying the triangle inequality [2, 30], we obtain

$$\left\|\theta_i^{t-1} - \theta_i^0\right\| \leq \sum_{j=1}^{t-1} \eta N_s \left\|\nabla \mathcal{L}_j(\theta_i^{j-1})\right\| \leq \eta N_s G(t-1), \tag{10}$$

where N_s is the number of epochs of the step-based update described above, and $G > 0$ is an upper bound on the Euclidean norm of the gradients. Therefore, we have

$$\|\Delta_t\| \leq L \cdot \eta N_s G(t-1). \tag{11}$$

Naturally, drawing on Theorem 3.1 and Eq. (11), the variation in update trajectories between the step-based and batch-based strategies can be expressed as follows:

$$\begin{aligned} \Delta_\theta &= \left\| -\eta N_s \sum_{t=1}^{T} \nabla_\theta \mathcal{L}_t(\theta_i^{t-1}) + \frac{\eta}{T} \sum_{j=1}^{N_b} \sum_{t=1}^{T} \nabla_\theta \mathcal{L}_t(\theta_i^{j-1}) \right\|, \\ &= \left\| \eta N_s \sum_{t=1}^{T} \left(\nabla_\theta \mathcal{L}_t(\theta_i^0) + \Delta_t\right) - \eta N_s \sum_{t=1}^{T} \nabla_\theta \mathcal{L}_t(\theta_i^0) \right\|, \\ &= \eta N_s \left\| \sum_{t=1}^{T} \Delta_t \right\| \leq \eta N_s \Big(\sum_{t=1}^{T} L \cdot \eta N_s G(t-1) \Big), \end{aligned} \tag{12}$$

where $\Delta_\theta = \left\|\theta_i^T - \theta_i^{N_b}\right\|$ represents the discrepancy in the cumulative update magnitudes between these two strategies. This quantity can be further bounded, as formalized in the following theorem.

Theorem 3.2 (Trajectory Update Error Bound). *Assume that the loss function $\mathcal{L}(\cdot)$ is differentiable with L-Lipschitz continuous gradients, and that the gradient norms remain uniformly stable throughout the update process. Here $L > 0$ denotes the smoothness constant, and $G > 0$ is an upper bound on the Euclidean norm of the gradients across all samples. Let η be a sufficiently small learning rate without decay. Therefore, for both step-based and batch-based update strategies, the trajectory difference after T steps is bounded by*

$$\begin{aligned} \left\|\theta_i^T - \theta_i^{N_b}\right\| &\leq \eta N_s \cdot L \eta N_s G \cdot \frac{T(T-1)}{2}, \\ &\leq \frac{L}{2} G \eta^2 N_s^2 T(T-1). \end{aligned} \tag{13}$$

Theorem 3.2 reveals the quantitative gap between step-based and batch-based update trajectories, showing that the discrepancy grows quadratically with respect to the number of steps, learning rate,

Algorithm 1 The Smooth-step Updating Strategy

Input: The examinee s_i; initial ability parameter $\theta_i^{'}$; interaction data $\{\mathcal{D}_i^1, \mathcal{D}_i^2, \ldots, \mathcal{D}_i^T\}$; learning rate η; step size k; local update epochs N
Output: Updated ability parameter θ_i^M

```
Initialize θ_i^0 ← θ_i'
M ← T/k                                   // number of update steps
for j = 1 to M do
    B_j ← {D_i^t}_{t=(j−1)k+1}^{jk}       // mini-batch in window j
    for epoch = 1 to N do
        Compute gradient: g ← (1/k) Σ_{b_t ∈ B_j} ∇_θ L_t(θ_i^{j−1}, b_t)
        Update parameter: θ_i^{j−1} ← θ_i^{j−1} − η · g
    end for
    Update: θ_i^j ← θ_i^{j−1}
end for
return θ_i^M
```

and step-based epoch count. This result provides a theoretical foundation for approximating the two strategies and enables fair comparisons in downstream tasks such as adaptive item selection.

3.4 The Smooth-step Updating Strategy

As previously discussed, given the update sensitivity of step-based strategy and the delayed responsiveness of batch-based strategy, we propose a smooth-step updating strategy to balance adaptivity and stability in the update process, as outlined in Algorithm 1, and visually demonstrated in Figure 1(c). In essence, the core idea is to set the update step size to k, so that each update is performed on a local batch of samples collected over a sliding window of size k, effectively forming a periodic mini-batch learning strategy. Formally, each update step can be defined as:

$$\theta_i^j = \theta_i^{j-1} - \eta N \cdot \frac{1}{k} \sum_{t=(j-1)k+1}^{jk} \nabla_\theta \mathcal{L}_t \left(\theta_i^{j-1}, \mathcal{D}_i^t\right), \tag{14}$$

where $t \in [(j-1)k+1,\ jk]$ denotes the sample indices within the j-th time window, and N represents the number of epochs used for parameter updates within each iteration. Accordingly, the overall update trajectory is given by:

$$\theta_i^M = \theta_i^0 - \eta N \cdot \frac{1}{k} \sum_{j=1}^{M} \sum_{t=(j-1)k+1}^{ik} \nabla_\theta \mathcal{L}_t \left(\theta_i^{j-1}, \mathcal{D}_i^t\right), \tag{15}$$

where $M = T/k$ denotes the number of mini-batch iterations. Then, following Eq. (9), the gradient perturbation term can be expanded as:

$$\nabla_\theta \mathcal{L}_j(\theta_i^{j-1}) = \nabla_\theta \mathcal{L}_j(\theta_i^0) + \delta_j, \tag{16}$$

where $\|\delta_j\| \le L \cdot \|\theta_i^{j-1} - \theta_i^0\| \le L\eta NG(j-1)$ provides an upper bound on the deviation introduced by using local parameter states in smooth-step updates. Leveraging this result and following the derivation in Theorem 3.2, we obtain the following bound on the discrepancy between the smooth-step and batch-based strategies:

$$\begin{aligned} \|\theta_i^M - \theta_i^{N_b}\| &\le \eta L \cdot \frac{N}{T} \sum_{j=1}^{M} \sum_{t=(j-1)k+1}^{jk} \left\|\theta_i^{j-1} - \theta_i^0\right\|, \\ &\le \eta L \cdot \frac{N}{M} \sum_{j=1}^{M} \left\|\theta_i^{j-1} - \theta_i^0\right\|, \\ &\le \eta L \cdot \frac{N}{M} \sum_{j=1}^{M} \eta NG(j-1), \\ &\le \frac{1}{2}\eta^2 N^2 GL(M-1), \end{aligned} \tag{17}$$

where N is regarded as an approximate value in both cases. The bound indicates that the deviation between smooth-step and batch-based updates scales linearly with the number of mini-batches M, and quadratically with both the learning rate η and the local update count N. The proposed smooth-step strategy, though based on stale parameters θ_i^{j-1}, offers advantages in practice. By enabling localized updates within each time window, it enhances computational efficiency and reduces communication overhead, while ensuring that the cumulative deviation remains theoretically bounded.

Moreover, the bound illustrates the impact of the mini-batch size k on update behavior. Specifically, when $k = 1$, the method reduces to step-based updates with an error bound of $O(\eta^2 N^2)$; when $k = T$, it corresponds to batch-based updates with zero error. As k increases, the deviation diminishes and stability improves, whereas smaller values of k allow for more responsive updates.

Table 1: The statistics of all datasets.

Statistics	ASSIST12	JUNYI
#Students	16,412	54,564
#Questions	3,906	565
#Knowledge concepts	119	30
#Interaction records	970,202	1,711,210
Avg. interactions per student	59.12	31.36
Avg. questions per concept	32.82	18.83

4 Experiments

4.1 Experimental Setting

4.1.1 ***Datasets.*** We conducted experiments on two real-world educational datasets with distinct characteristics, namely ASSIST12 [25] and JUNYI [5], to investigate the performance differences among various update strategies. The statistics of the datasets are summarized in Table 1, and the corresponding details are described below:

- **ASSIST12** [25]: ASSIST12 is a dataset collected from the ASSISTments online educational tutoring system, containing rich student interactions with a variety of exercises. It records detailed information on student responses, exercise attempts, and timestamps, and has been widely used in adaptive testing tasks.
- **JUNYI** [5]: The JUNYI Academy dataset, provided by Junyi online learning platform, contains large-scale student interaction records across various subjects, including responses, timestamps, and exercise attempts. It also includes prerequisite knowledge

Table 2: Performance comparison of SUAT and all baseline update methods on the ASSIST12 and JUNYI datasets in terms of ACC and AUC across different testing lengths. Bold highlights the best performance, and underline denotes the runner-up.

CDM		MIRT				NCD			
Dataset/Metric		ASSIST12 / ACC/AUC(%) ↑							
QSM	Method	Metric@5	Metric@10	Metric@20	Metric@Avg.	Metric@5	Metric@10	Metric@20	Metric@Avg.
MKLI	STEP	70.60/67.17	70.71/67.73	70.94/68.44	70.75/67.78	66.56/62.08	67.09/62.34	68.98/63.41	67.54/62.61
	BATCH	67.86/60.42	69.26/63.20	70.74/66.28	69.32/63.30	66.87/62.33	68.05/62.49	69.17/64.09	68.03/62.97
	In-Batch	70.62/66.59	70.85/67.56	71.27/68.46	70.85/67.14	**67.67/64.63**	66.81/62.35	68.23/63.42	67.57/63.47
	SUAT	**70.67/67.33**	**70.93/67.91**	**71.32/68.73**	**70.97/67.99**	67.44/63.96	**68.11/62.57**	**69.43/64.38**	**68.32/63.63**
MAAT	STEP	70.27/**66.35**	70.48/65.40	71.11/66.37	70.62/66.04	66.37/63.39	65.53/61.74	69.47/63.62	67.12/62.91
	BATCH	66.76/59.42	69.25/60.73	70.06/63.19	68.34/61.17	65.31/61.58	68.49/62.76	69.21/64.73	67.67/63.02
	In-Batch	70.38/65.08	70.81/65.11	71.25/66.22	70.82/65.47	**67.18**/62.30	65.39/61.98	68.46/63.82	67.01/62.70
	SUAT	**70.61**/66.26	**70.86/65.74**	**71.32/66.50**	**70.94/66.16**	66.43/**63.58**	**68.75/63.01**	**69.98/64.85**	**68.71/63.48**
BOBCAT	STEP	**70.40**/66.72	70.42/**67.24**	70.46/67.50	70.42/67.15	67.24/**64.63**	**67.49**/62.31	67.82/63.69	67.51/63.54
	BATCH	66.31/59.21	67.10/61.36	68.72/64.74	67.37/61.78	67.05/62.17	67.14/63.52	67.83/66.03	67.34/63.90
	In-Batch	70.08/65.73	70.36/66.15	70.51/67.62	70.31/66.50	66.52/63.31	67.06/62.11	67.50/63.68	67.02/63.03
	SUAT	70.36/**66.80**	**70.44**/66.76	**70.65/68.01**	**70.45/67.19**	**67.27**/63.38	67.38/**63.55**	**67.97/66.45**	**67.54/64.46**
Dataset/Metric		JUNYI / ACC/AUC(%) ↑							
QSM	Method	Metric@5	Metric@10	Metric@20	Metric@Avg.	Metric@5	Metric@10	Metric@20	Metric@Avg.
MKLI	STEP	78.54/79.89	83.33/83.38	87.35/86.63	83.07/83.30	67.39/**79.75**	70.85/79.27	78.92/81.01	72.38/**80.02**
	BATCH	85.97/80.97	87.32/85.85	88.47/88.15	87.25/84.99	68.02/78.18	74.65/79.25	81.31/81.07	74.66/79.50
	In-Batch	79.05/78.91	86.39/85.90	88.02/88.21	84.48/84.34	67.23/79.07	71.84/79.12	79.56/80.99	72.87/79.72
	SUAT	**86.45/81.48**	**87.51/86.45**	**88.86/88.72**	**87.61/85.55**	**68.34**/78.26	**74.88/79.46**	**81.52/81.32**	**74.92**/79.69
MAAT	STEP	72.09/**82.73**	77.33/85.43	82.13/87.92	77.18/**85.36**	66.32/79.02	68.57/78.31	79.71/82.43	71.53/79.92
	BATCH	81.20/81.23	85.44/85.39	87.54/88.11	84.72/84.91	66.57/78.47	77.44/81.07	79.16/82.34	74.39/80.62
	In-Batch	75.91/80.34	84.80/84.26	87.58/**88.23**	82.76/84.27	66.49/79.06	70.08/78.93	78.04/82.53	71.54/80.17
	SUAT	**81.52**/81.34	**85.53/85.68**	**87.67**/87.90	**84.91**/84.97	**66.78/79.45**	**77.72/81.49**	**79.97/82.85**	**74.82/81.26**
BOBCAT	STEP	79.40/84.19	85.28/86.71	87.53/87.82	84.07/86.24	73.38/80.52	77.40/78.57	80.48/78.61	77.09/79.23
	BATCH	85.03/80.27	86.92/81.70	87.79/85.20	85.58/82.39	75.53/80.03	78.39/78.02	82.36/78.89	78.76/78.98
	In-Batch	81.14/84.59	86.91/86.88	88.12/88.36	86.04/86.61	74.77/80.46	78.35/78.60	81.90/78.10	78.34/79.05
	SUAT	**85.51/85.28**	**87.20/87.05**	**88.51/88.73**	**86.91/86.92**	**75.64/80.70**	**78.50/79.04**	**82.45/79.84**	**78.86/79.86**

graphs, making it widely used for knowledge tracing, learning trajectory analysis, and personalized educational research.

4.1.2 ***Baseline Methods.*** To comprehensively evaluate the performance of different update strategies under various question selection algorithms, we carefully designed and configured the experimental settings. Specifically, we selected three question selection algorithms, MKLI [3], MAAT [1], and BOBCAT [12], covering both rule-based and data-driven approaches. Meanwhile, we compared three update strategies: in addition to the step-based (**STEP**) and batch-based (**BATCH**) strategies discussed in the paper, we additionally include a variant of the batch-based strategy, termed **In-Batch**, which performs iterative stepwise updates within each mini-batch of samples. The details of the three selection methods are described as follows, including their core ideas and procedures.

- **MKLI** [3]: It identifies exercises based on the highest moving average of Kullback-Leibler information, enabling efficient evaluation while accommodating multidimensional student abilities.
- **MAAT** [1]: It is an active learning-based method that incorporates quality, diversity, and importance modules, and selects exercises by estimating the Expected Model Change (EMC) within the cognitive diagnosis model (CDM).
- **BOBCAT** [12]: It formulates CAT as a bi-level optimization problem, in which the item selector is adaptively learned from data, thereby overcoming the rigidity and strong assumptions inherent in traditional rule-based or IRT-based models.

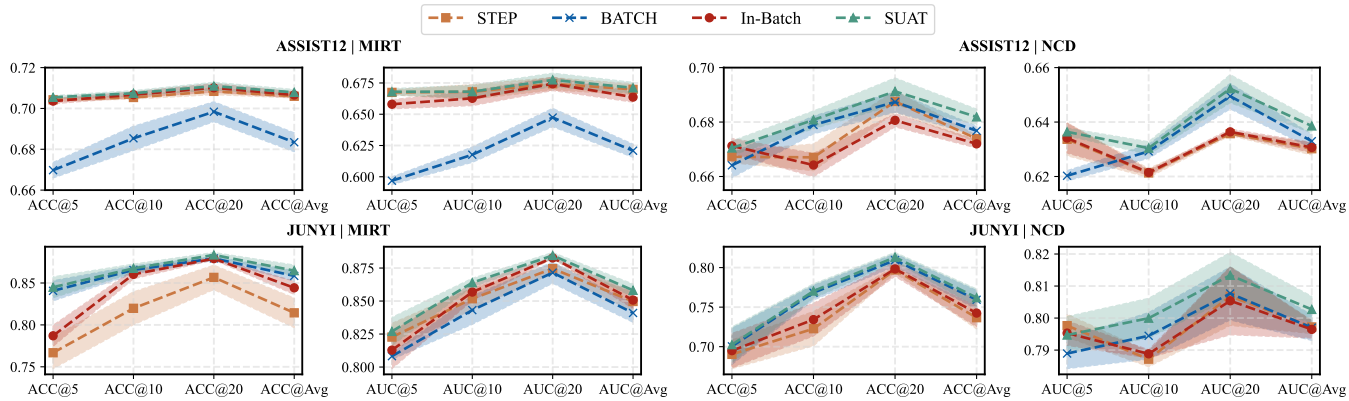


Figure 2: The error band plot illustrating the performance of SUAT and all baseline methods on the ASSIST12 and JUNYI datasets across different lengths. The dashed line denotes the mean, and the transparent range indicates the standard deviation.

4.1.3 ***Evaluation Metrics.*** Following the evaluation protocol of previous CAT methods [1, 56], we used the area under the ROC curve (AUC) and accuracy (ACC) to assess model performance in the experiments. To comprehensively evaluate performance across different testing lengths, we report results at lengths of 5, 10, and 20, as well as their average for comparison, denoted as Metric@5, Metric@10, Metric@20, and Metric@Avg., respectively.

4.1.4 ***Parameter Settings.*** Following prior work [56], we adopt MIRT [28] and NCD [36] as the foundational ability assessment modules. During the pre-training of $\mathcal{M}_a$, student and question representations are initialized with dimensions matching the number of knowledge concepts. In the adaptive testing assessment phase, the question selection module $\mathcal{M}_s$ is configured with consistent settings: the learning rates for MIRT and NCD are set to 0.02 and 0.005, respectively. The number of updating per step N_s is fixed at 5 in the step-based strategy, while the step size in per batch k and the update epochs N are both searched over the candidate set {1, 2, 3, 4, 5}. All methods are implemented in Python using PyTorch, with the Adam optimizer employed for parameters training.

4.2 Experimental Results and Discussion

Table 2 presents the model performance of four update mechanisms (including our proposed SUAT) on the ASSIST12 and JUNYI datasets. The best result for each metric is highlighted in bold, while the runner-up is underlined. It is worth noting that all results were obtained through multiple rounds of experimental validation, and the reported values represent averaged outcomes to mitigate random fluctuations. Based on these results, we make the following observations: **(1)** Step-based and batch-based update strategies demonstrate comparable performance in most cases, supporting the conclusion of Theorem 3.1—that when $N_b \approx T \cdot N_s$, the update strengths of the two strategies are approximately equivalent. However, notable performance gaps can still arise under certain configurations. For example, under the MAAT question selection method on the JUNYI dataset, the average ACC difference between the two strategies reaches approximately 7.5%. Such discrepancies are often attributed to biases in the question selection strategy and the inherent characteristics of the dataset. Specifically, MAAT emphasizes the diversity of knowledge concepts in selected questions, and due to differences in update granularity, these questions exert varying levels of influence on the two strategies, leading to divergent outcomes; **(2)** Batch-based and In-Batch update methods exhibit highly similar performance, differing only in whether the overall gradient error across each batch is considered during updates. On the ASSIST12 dataset, In-Batch methods generally outperform batch-based ones, particularly under the BOBCAT question selection strategy, with an average accuracy (ACC) improvement of approximately 2.5%. This suggests that in datasets with denser question features, more fine-grained updates tailored to batch data may lead to improved performance; **(3)** The overall performance advantage of the proposed SUAT method is evident; however, it is reasonable that some of its results are suboptimal. This can be attributed to the fact that in datasets exhibiting substantial variations in feature distributions, and under different question selection strategies with distinct emphases, the advantages of step-based and batch-based methods may not necessarily complement each other effectively.

To further illustrate the performance variability of different update strategies across various selection methods, we visualized the results from Table 2 using error band diagrams, as shown in Figure 2. The colored lines denote the mean performance of each ability update method, while the shaded areas represent the corresponding standard deviation ranges. The x-axis indicates performance metrics at different test lengths-5, 10, and 20-as well as their averaged values. Overall, the following observations can be made: **(1)** Step-based and batch-based methods demonstrate distinct differences and limitations across datasets under the MIRT diagnosis model. For instance, in the ASSIST12 dataset, batch-based methods exhibit notable drawbacks, with performance gaps reaching up to 3% compared to the best-performing approaches, even in shorter evaluations. Conversely, in the JUNYI dataset, step-based methods show substantial performance declines, with accuracy differences approaching 10%. These results suggest considerable instability in the ability update processes of both methods across different datasets; **(2)** It is evident that all update strategies based on the NCD model display considerably greater performance volatility compared to those based on the MIRT model. This is likely due to

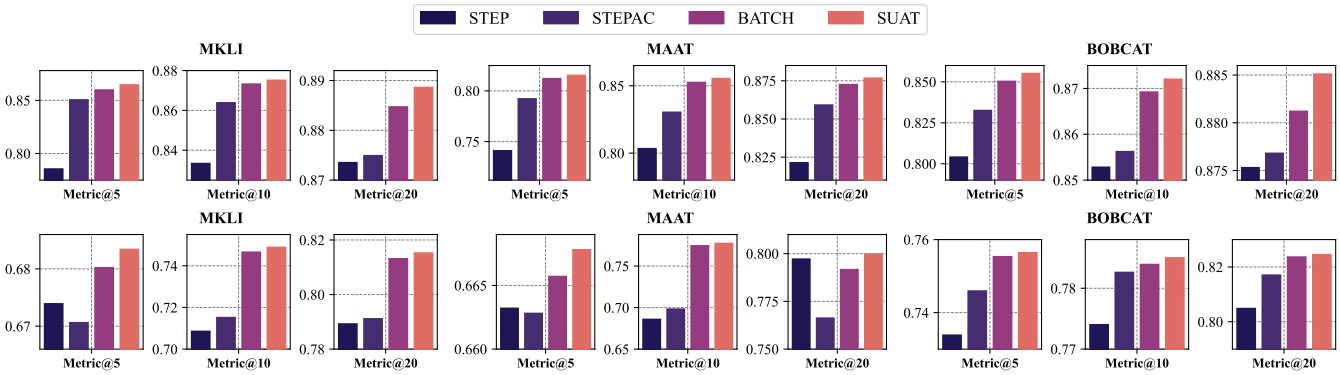


Figure 3: Performance comparison of four update strategies incorporating the step-based accumulation update method on the JUNYI dataset in terms of ACC. The top and bottom panels show results under MIRT and NCD assessment models, respectively.

Table 3: Parameter analysis of the step-size k in the smooth-step updating strategy on the JUNYI dataset.

Params		$k=1$	$k=2$	$k=3$	$k=4$	$k=5$
MAAT	@5	**81.52**/**81.34**	80.38/78.85	79.68/77.17	78.28/77.22	79.12/74.09
	@10	85.53/**85.68**	**85.61**/81.61	83.51/79.75	84.71/79.12	83.35/78.31
	@20	**87.67**/87.90	87.55/86.06	86.88/**87.96**	86.70/84.57	86.68/83.98
	Avg.	**84.91**/**84.97**	84.51/82.17	83.35/81.62	83.23/80.30	83.05/78.79
BOBCAT	@5	**85.51**/**85.28**	85.24/84.69	84.84/84.17	83.35/82.91	83.41/82.92
	@10	87.20/**87.05**	**87.21**/86.72	86.88/85.69	86.63/85.24	86.24/84.98
	@20	88.51/**88.73**	88.10/88.24	88.09/87.86	**88.54**/87.70	88.05/87.20
	Avg.	**86.91**/**86.92**	86.85/86.55	86.61/85.91	86.17/85.28	85.91/85.03

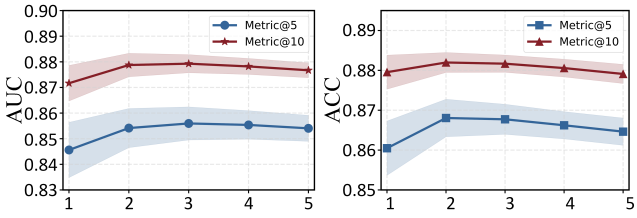


Figure 4: Parameter analysis of the update epoch N in the smooth-step updating strategy on the JUNYI dataset.

the NCD approach introducing multidimensional capability mappings, which increase the complexity of the representation space. As a result, the learning of test takers' capability updates becomes more sensitive to variations in the question selection methods.

In addition, to evaluate the effectiveness of the incremental update strategy [32], we introduced a step-based accumulation update method on the JUNYI dataset for performance comparison. Specifically, this method builds upon the step-based approach (termed **STEPAC**), but at each update step, it incorporates all accumulated data from previous steps. As shown in Figure 3, it yields significant performance improvements across various metrics compared to the standard step-based update method. This demonstrates that

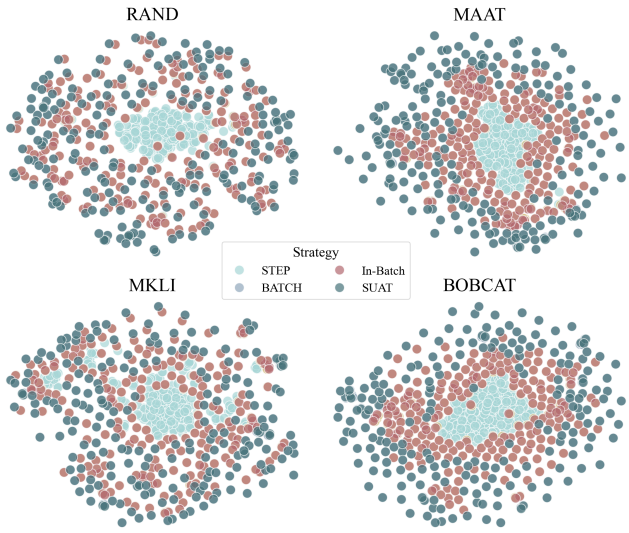


Figure 5: Visualization of the final student abilities obtained with different update strategies on the ASSIST12 dataset.

the incremental update strategy, by repeatedly leveraging prior information and making more comprehensive use of the data, naturally offers advantages in updating and refining students' ability parameters. However, this also comes at the cost of increased computational overhead. Meanwhile, a noticeable performance gap can be observed between the two step-based update methods and the batch-based method, further demonstrating the advantage of SUAT.

4.3 Trajectory Parameter Analysis

In this section, we performed a comprehensive update trajectory parameter analysis to assess the impact of hyper-parameters, including the step-size k and the update epoch N. Due to space limitation, we mainly show the results for the JUNYI dataset. Specifically, we first set the value of k to {1, 2, 3, 4, 5}. Table 3 primarily presents the results under the MAAT and BOBCAT selection models. It can be observed that as the step-size k increases, the overall performance of the model tends to decline. In most cases, setting $k = 1$ or $k = 2$

yields better results, indicating that even within batch-based update strategies, finer-grained step sizes can offer clear performance benefits. In addition, we conducted a sensitivity analysis of the update epoch N in SUAT, evaluating values in the range {1, 2, 3, 4, 5}. As shown in Figure 4, variations in N have a limited impact on the model's overall performance. Notably, the best results are achieved when $N = 2$ or $N = 3$, suggesting an optimal trade-off between effectiveness and computational efficiency. Interestingly, unlike the trends observed in step-based and batch-based methods, SUAT's performance does not exhibit a linear improvement as N increases.

4.4 Case Study

To better understand how different update strategies influence the final student ability representations after the testing process, we conducted a visualization case study on the ASSIST12 dataset by sampling the final embedding vectors of 200 students. Figure 5 presents the t-SNE visualizations of student representations obtained under four different ability update methods across four question selection strategies. Notably, representations produced by the step-based method appear more tightly clustered, indicating that finer-grained, stepwise updates facilitate faster convergence and more compact embedding structures. In contrast, the batch-based method generates a more uniformly distributed embedding space, reflecting the smoothing effect of aggregating updates over entire batches. Furthermore, due to the methodological similarities between the batch-based and in-batch strategies, their resulting embeddings tend to overlap, highlighting the partial retention of stepwise dynamics in the in-batch approach. In fact, this intuitive visualization can also assist in exploring and selecting different parameter update strategies for downstream applications, such as exercise recommendation and learning path planning.

5 Conclusion

In this paper, we began by disentangling the dynamics of step-based and batch-based update strategies, analyzing their similarities through the lens of update intensity and quantifying their divergence via theoretical error bounds. To strike a balance between adaptability and stability, we introduced **SUAT**, a **S**mooth-step **U**pdating strategy for more effective **A**daptive **T**esting. Through comprehensive experiments on two real-world educational datasets of varying scales, we empirically validate our theoretical findings and reveal distinct behavioral patterns across different update mechanisms. These results not only provide practical insights into the strengths and limitations of each strategy but also highlight their implications for real-world adaptive testing systems, informing the design of more effective ability estimation procedures, exercise recommendation policies, and personalized learning path planning.

Acknowledgments

This work was supported in part by the National Natural Science Foundation of China (No.U21A20512, No.62107001, No.62302010), in part by China Postdoctoral Science Foundation (No.2023M740015), and in part by the Postdoctoral Fellowship Program (Grade B) of China Postdoctoral Science Foundation (No.GZB20240002).

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